

Unsupervised learning with Dimensionality reduction and Clustering

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• Abstract

This project demonstrates the application of Principal Component Analysis (PCA) and K-Means Clustering on the handwritten digits dataset available in Scikit-Learn. The dataset contains 1797 images of handwritten digits represented by 64 pixel features. PCA was used to reduce the dimensionality of the dataset from 64 dimensions to 2 dimensions for visualization and efficient clustering. K-Means clustering was then applied to group similar digit images into 10 clusters. The cluster assignments were mapped to actual digit labels using the mode of the true labels present in each cluster. The performance of the clustering model was evaluated using accuracy score, confusion matrix, and classification report. The obtained accuracy was approximately 56.65%. The results show that unsupervised clustering can identify several digit groups successfully, although some digits were heavily misclassified due to dimensionality reduction and the unsupervised nature of the algorithm. This project provides practical exposure to machine learning workflows involving data preprocessing, dimensionality reduction, clustering, and model evaluation.

• Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make decisions without being explicitly programmed. One important problem in machine learning is image recognition, where computers attempt to identify objects or patterns present in images.

In this project, handwritten digit images from the Scikit-Learn digits dataset were analyzed. Each image consists of 8×8 pixels and is represented by 64 numerical features. Since working directly with high-dimensional data can be computationally expensive and difficult to visualize, Principal Component Analysis (PCA) was used to reduce the dimensionality of the data to two principal components.

After dimensionality reduction, K-Means Clustering was applied to group similar digit images into ten clusters. The resulting clusters were analyzed and mapped to actual digit labels. The performance of the clustering model was evaluated using standard machine learning metrics such as accuracy score, confusion matrix, precision, recall, and F1-score.

Topics covered during internship training included Python Programming Basics, NumPy, Pandas, Data , variables ,lists and loop ,Data Structures ,class ,Functions ,OOPS. Introduction to Machine Learning(Overview),LLM fundamentals and Communication Skills.

• Project Objective

- To understand the structure and characteristics of the handwritten digits dataset.
- To apply Principal Component Analysis (PCA) for dimensionality reduction.

- To perform unsupervised learning using the K-Means clustering algorithm.
- To visualize clustered handwritten digit data in two-dimensional space.
- To evaluate clustering performance using accuracy score, confusion matrix, and classification report.

● Methodology

- **Step 1:** Import the required libraries such as NumPy, Matplotlib, Scikit-Learn, and SciPy.
- **Step 2:** Load the handwritten digits dataset using the `load_digits()` function from Scikit-Learn.
- **Step 3:** Explore the dataset and verify its structure. The dataset contains 1797 handwritten digit images represented by 64 pixel features.
- **Step 4:** Apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset from 64 dimensions to 2 dimensions while preserving the maximum possible variance.
- **Step 5:** Apply K-Means clustering with 10 clusters on the reduced two-dimensional dataset.
- **Step 6:** Obtain cluster labels for all 1797 observations using the `fit_predict()` method of the K-Means algorithm.
- **Step 7:** For each cluster, determine the most frequently occurring actual digit using the `mode()` function. This creates a mapping between cluster labels and digit labels.
- **Step 8:** Convert the cluster labels into predicted digit labels using the cluster-to-digit mapping. These predicted labels represent the model's estimated digit for each image.
- **Step 9:** Compare the predicted digit labels with the actual digit labels stored in `digits.target`. The `digits.target` array contains the true digit (0–9) corresponding to each handwritten image in the dataset.
- **Step 10:** Calculate the clustering accuracy using the `accuracy_score()` function. Accuracy is computed as the proportion of correctly predicted digit labels to the total number of observations.
- **Step 11:** Generate a confusion matrix to analyze the number of correct and incorrect predictions for each digit class.
- **Step 12:** Generate a classification report containing precision, recall, and F1-score values for each digit class to evaluate the performance of the clustering model.
- **Tools Used:**
 - ❖ Python
 - ❖ Jupyter Notebook / Google Colab
 - ❖ Scikit-Learn
 - ❖ NumPy
 - ❖ Matplotlib
 - ❖ SciPy
 - ❖ GitHub

The Github link has been shared below:

https://github.com/yaminishaw6-ai/10-Unsupervised_Learning_with_Dimensionality_Reduction_and_Clustering_summer_2026/upload/main

• Data Analysis and Results

- **DATASET INFORMATION:** The handwritten digits dataset was loaded using the Scikit-Learn library. The dataset contains 1797 handwritten digit images. Each image is represented by 64 pixel intensity values corresponding to an 8×8 image.

Dataset Shape:

Number of observations: 1797 ,Number of features: 64

```
[66] ... (1797, 64)
```

- **PRINCIPAL COMPONENT ANALYSIS(PCA):** Since the original dataset consists of 64 dimensions, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset to two dimensions. This reduction helps in visualization and improves the efficiency of clustering.

Reduced Dataset Shape:

Number of observations: 1797 ,Number of principal components: 2

```
] (1797, 2)
```

- **K –MEANS CLUSTERING RESULTS:** K-Means clustering was applied on the PCA-reduced dataset with the number of clusters set to 10. Each cluster represents a group of similar handwritten digit images. The first 10 cluster labels of the first 10 digits were printed as follows

```
[83] ... [9 4 5 6 2 6 2 8 3 3]
```

The cluster labels generated by K-Means were mapped to actual digit labels using the Mode of the true labels present in each cluster.

```
{0: np.int64(2), 1: np.int64(0), 2: np.int64(4), 3: np.int64(8),
```

Empty markdown cell, double-click or press enter to edit.

```
4(8), 4: np.int64(1), 5: np.int64(8), 6: np.int64(9), 7: np.int64(6),
```

```
8: np.int64(7), 9: np.int64(0)}
```

The output shows the mapping between cluster labels and actual digit labels.

Each cluster was assigned the digit that appeared most frequently within that cluster, allowing the clustering results to be interpreted as digit predictions.

- **GENERATION OF PREDICTED LABELS :**Using the cluster-to-digit mapping obtained in the previous step, each cluster label was converted into a predicted digit label. For example, if a cluster was mapped to digit 8, then all observations belonging to that cluster were assigned the predicted label 8.

The first few predicted labels obtained were:

```
1 [0 1 8 9 4 9 4 7 8 8]
```

These predicted labels were then compared with the actual digit labels stored in digits.

Target to evaluate the performance of the clustering model

- **MODEL EVALUATION:**The predicted digit labels obtained from clustering were compared with the actual digit labels stored in digits.target.

Accuracy Score:

The model achieved an accuracy of approximately 56.65%.

```
1 0.5664997217584864
```

- **CONFUSION MATRIX ANALYSIS :** A confusion matrix was generated to analyze the classification performance of the clustering model.

```
1 [[167  0  0  0  1  0  7  0  3  0]
  [ 2 97  0  0  4  0  0  0 11 68  0]
  [ 0  3 114  0  0  0  0  0 10 47  3]
  [ 2  0  63  0  0  0  0  0 13 15 90]
  [ 0 11  0  0 138  0 27  3  2  0]
  [14  6  4  0  2  0  4 23 104 25]
  [22  0  0  0 21  0 136  0  2  0]
  [ 0 35  0  0  1  0  0 115 28  0]
  [ 0 10  3  0  1  0  0 24 136  0]
  [ 4  1  8  0  0  0  1  9 42 115]]
```

From the confusion matrix, it was observed that digits 0, 4, 6, 7, 8, and 9 were identified relatively well. However, digits 3 and 5 were frequently misclassified into other clusters, resulting in reduced overall accuracy.

- **CLASSIFICATION REPORT:**The classification report was generated to evaluate precision, recall, and F1-score for each digit class.

	precision	recall	f1-score	support
0	0.79	0.94	0.86	178
1	0.60	0.53	0.56	182
2	0.59	0.64	0.62	177
3	0.00	0.00	0.00	183
4	0.82	0.76	0.79	181
5	0.00	0.00	0.00	182
6	0.78	0.75	0.76	181
7	0.55	0.64	0.59	179
8	0.30	0.78	0.44	174
9	0.49	0.64	0.56	180
accuracy			0.57	1797
macro avg	0.49	0.57	0.52	1797
weighted avg	0.49	0.57	0.52	1797

The results indicate that digit classes 0, 4, and 6 achieved comparatively high precision and recall values. Digit classes 3 and 5 obtained very low performance scores because the clustering algorithm failed to form distinct clusters for these digits. The overall model accuracy was approximately 56.65%.

• Conclusion

This project successfully demonstrated the use of Principal Component Analysis (PCA) and K-Means Clustering on handwritten digit data. PCA reduced the original 64-dimensional dataset to two dimensions, enabling easier visualization and clustering. K-Means clustering grouped similar digit images into ten clusters, and these clusters were mapped to actual digit labels for evaluation. The model achieved an accuracy of approximately 56.65%. The results indicate that while clustering can successfully identify several digit groups, some digits remain difficult to separate after dimensionality reduction. Future work may involve using additional principal components or supervised machine learning algorithms such as K-Nearest Neighbors (KNN), Support Vector Machines (SVM), or Random Forests to improve classification performance.

• APPENDICES

▪ References

Scikit-Learn Documentation

PCA Documentation

K-Means Documentation

Python Documentation

- GitHub Repository Link : https://github.com/yaminishaw6-ai/10-Unsupervised_Learning_with_Dimensionality_Reduction_and_Clustering_summer_2026/upload/main
- A copy of the project report :
<https://drive.google.com/drive/folders/15nwEDdEcroNe73h1liSSzvsKmTZ05JTx>
- A copy of the video recording :
- <https://drive.google.com/drive/folders/15nwEDdEcroNe73h1liSSzvsKmTZ05JTx>